



Opportunities and challenges of using circulating tumor DNA to predict lung cancer immunotherapy efficacy

Shanshan Li^{1,2} · Ting Yuan² · Jing Yuan³ · Bo Zhu^{2,4} · Degao Chen^{2,4}

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Abstract

Immune checkpoint inhibitors (ICIs), particularly anti-programmed death 1 (PD-1)/ programmed death ligand 1 (PD-L1) antibodies, have led to significant progress in lung cancer treatment. However, only a minority of patients have responses to these therapies. Detecting peripheral blood of circulating tumor DNA (ctDNA) allows minimally invasive diagnosis, characterization, and monitoring of lung cancer. ctDNA has potential to be a prognostic biomarker and a predictor of the response to ICI therapy since it can indicate the genomic status and tumor burden. Recent studies on lung cancer have shown that pretreatment ctDNA analysis can detect residual proliferative disease in the adjuvant immunotherapy setting and evaluate tumor burden in patients with metastatic disease. Early ctDNA dynamics can not only predict the clinical outcome of ICI therapy but also help distinguish between pseudoprogression and real progression. Furthermore, in addition to quantitative assessment, ctDNA can also detect genetic predictors of response to ICI therapy. However, barriers still exist in the application of ctDNA analysis in clinical lung cancer treatment. The predictive value of ctDNA in lung cancer immunotherapy requires further identification and resolution of these challenges. This review aims to summarize the existing data of ctDNA analysis in patients receiving immunotherapy for lung cancer, understand the limitations of clinical treatment, and discuss future research directions.

Keywords Circulating tumor DNA · Immune checkpoint inhibitors · bTMB · Mutation · Lung cancer

Introduction

With an expected 2.2 million new cancer cases and 1.8 million deaths in 2020, lung cancer is one of the most common malignancies to get a diagnosis and is the leading cause of cancer-related death worldwide (Sung et al. 2021). Non-small cell lung cancer (NSCLC) and small cell lung cancer (SCLC) are the two main forms of lung cancer. Most patients with lung cancer are already in the advanced stage at the time of diagnosis since the clinical symptoms are not obvious, and the 5-year survival rate is just 10–20% (Sung et al. 2021). Immunotherapy using ICIs has led to rapid progress in lung cancer treatment over the past decade. ICIs represented by anti-programmed death 1 (PD-1)/programmed death ligand 1 (PD-L1) have significantly improved clinical outcomes in patients with extensive stage SCLC and advanced NSCLC (Borghaei et al. 2015; Reck et al. 2016; Gandhi et al. 2018; Goldman et al. 2021; Liu et al. 2021). Unfortunately, only a minority of patients respond to ICIs. In addition, patients treated with ICIs develop serious immune-related adverse events, such as pneumonia and

Shanshan Li and Ting Yuan contributed equally to this work.

✉ Bo Zhu
bo.zhu@tmmu.edu.cn

✉ Degao Chen
degaochen@tmmu.edu.cn

¹ School of Pharmacy and Bioengineering, Chongqing University of Technology, Chongqing 400054, China

² Institute of Cancer, Xinqiao Hospital, Third Military Medical University, Chongqing 400037, China

³ Center for Joint Surgery, Department of Orthopedic Surgery, The Second Affiliated Hospital of Chongqing Medical University, Chongqing 400010, China

⁴ Chongqing Key Laboratory of Immunotherapy, Xinqiao Hospital, Third Military Medical University, Chongqing 400037, China

hepatitis (Berti et al. 2021). Therefore, accurate biomarkers are needed to screen patients with lung cancer to determine who will benefit from ICIs.

Several randomized controlled trials have confirmed that PD-L1 expression and tumor mutation burden (TMB) predict ICI therapy response in NSCLC. However, they still face several challenges in clinical practice. First, PD-L1 expression and TMB are inaccurate as biomarkers of immunotherapy response. According to some research, patients with high expression of PD-L1 or a high TMB do not respond to ICIs, whereas those with low expression or a low TMB may respond (Carbone et al. 2017; Garassino et al. 2018; Reck et al. 2019; Hellmann et al. 2019; Fricke et al. 2020). Second, the detection of PD-L1 levels based on immunohistochemistry varies based on the platform and antibody used, and therefore they cannot be used to guide clinical treatment strategies (Topalian et al. 2016; Hirsch et al. 2017). The initial detection strategy for TMB was whole exome sequencing (WES) (Garofalo et al. 2016), which was gradually replaced by next-generation sequencing because of the cost and time required; however, only two gene panels, Foundation One CDX and MSK-IMPACT, have been approved by the US Food and Drug Administration (FDA) (Duffy and Crown 2019). In addition, there is temporal and spatial heterogeneity in PD-L1 expression (Mansfield et al. 2016; Liu et al. 2018; Munari et al. 2018), and as a continuous variable, TMB lacks an optimal cutoff point that can be applied to all tumor types (Samstein et al. 2019). Therefore, developing additional predictive biomarkers to assess the clinical efficacy of immunotherapy in lung cancer is imperative.

Recently, liquid biopsy, an emerging detection technology, has made increasing progress in clinical practice (Ignatiadis et al. 2021). Liquid biopsy not only obtains tumor samples in a minimally invasive manner but also provides a more acceptable, safe, and easy-to-repeat option for monitoring tumor treatment response than solid tumor biopsy (Stadler et al. 2022). Through liquid biopsy, the peripheral blood and other body fluids of tumor patients can be sampled to analyze circulating tumor cells, circulating tumor DNA (ctDNA), exosomes, and other analytes. ctDNA has high potential as a biomarker as it not only retains tumor-specific genetic and epigenetic characteristics but also has a short half-life (Diehl et al. 2008), making it ideal for rapidly monitoring tumor changes over time.

In this review, we present the different clinical applications of ctDNA analysis in the context of lung cancer immunotherapy. We further discuss the potential of ctDNA quantification before and during ICI therapy, examine its advantages and limitations as a tumor biomarker, and identify patients who respond to immunotherapy. Finally, we consider the genetic predictors associated with ICI response

to ctDNA and highlight the value of clinical decision-making based on ctDNA in lung cancer immunotherapy.

Pretreatment levels of ctDNA for predicting the response to ICIs in lung cancer

Predicting response to adjuvant/neoadjuvant immunotherapy

Neoadjuvant immunotherapy is currently administered to patients with resectable lung cancer. Neoadjuvant chemotherapy plus nivolumab was shown to be effective for resectable NSCLC in the NADIM trial; however, surrogate endpoints that can accurately predict long-term survival are still lacking (Provencio et al. 2020, 2022). Using the OncoPrint Pan-Cancer Cell-Free Assay, Provencio et al. detected ctDNA in 43 patients, defined it as the sum of all mutant allele fractions (MAF), and selected 1% MAF as a cutoff to predict response to therapy. The results showed that neither TMB nor PD-L1 expression could predict long-term survival, but patients with low pretreatment ctDNA levels had significantly better overall survival (OS) and progression-free survival (PFS) (Provencio et al. 2022). Additionally, clinical responses based on the RECIST v1.1 could not predict survival outcomes, suggesting that pretreatment ctDNA levels were superior to conventional imaging in evaluating prognosis (Provencio et al. 2022). More research is required to determine the optimal cutoffs and to explore and verify the prognostic significance of pretreatment ctDNA levels in neoadjuvant therapy.

ICIs have also been incorporated into adjuvant therapeutic strategies, offering an additional course of treatment for lung cancer patients who have received resection (Felip et al. 2021; O'Brien et al. 2022). However, the degree of treatment benefit should be weighed against the risk of immune-related adverse events in clinical practice. Thus, using biomarker strategies to determine patients at high risk of immune-related adverse events is necessary. In patients with early-stage disease, ctDNA analysis after definitive local therapy can be used to detect minimal residual disease (MRD), which helps identify patients at high risk of disease recurrence, and therefore those who are most likely to require adjuvant immunotherapy (Chaudhuri et al. 2017; Abbosh et al. 2018). Moding et al. used CAPP-Seq ctDNA to analyze plasma samples from patients receiving definitive chemoradiation therapy (CRT) for locally advanced NSCLC and found that ctDNA-based MRD detection had a high predictive value for the risk of disease progression after CRT, and could be detected earlier than radiographic progression (Moding et al. 2020). Patients with ctDNA detected only before consolidation immunotherapy had

better PFS than those with ctDNA detected before and during consolidation immunotherapy, which also indicates that patients with ctDNA MRD detected after CRT could selectively benefit from consolidation immunotherapy (Moding et al. 2020).

In clinical practice, ctDNA MRD may serve as a biomarker for predicting adjuvant immunotherapy responses after radical treatment. Randomized clinical trials, including two ongoing clinical trials, have prospectively evaluated the predictive value of ctDNA analysis for adjuvant immunotherapy. The Phase III MERMAID-1 clinical trial is assessing ctDNA analysis on plasma samples of postoperative patients to determine MRD status and has randomly assigned MRD-positive NSCLC patients to durvalumab or placebo with chemotherapy (NCT04385368) (Peters et al. 2021). The MERMAID-2 clinical trial randomly assigned patients with NSCLC who were evaluated as MRD-positive after curative-intent therapy to receive durvalumab or placebo (NCT04642469) (Spigel et al. 2021).

Predicting response to ICIs in metastatic lung cancer

The vast majority of cancer-related deaths are owing to disease progression, in which metastasis leads to impairment of important organ functions (Gupta and Massagué 2006; Chaffer and Weinberg 2011). In all cancer types, patients with metastatic cancer have higher ctDNA levels than patients with early-stage disease (Phallen et al. 2017). Therefore, ctDNA has been studied as a candidate predictive biomarker in several immunotherapy cohorts (Table 1).

Cabel et al. enrolled 10 patients with metastatic NSCLC treated with nivolumab and observed associations between baseline ctDNA levels and outcomes (Cabel et al. 2017). Undetectable baseline ctDNA levels were associated with improved OS but not PFS. A study involving 100 patients with advanced lung adenocarcinoma who were treated with ICIs revealed that patients with low baseline ctDNA levels had better OS and PFS than those with high baseline ctDNA levels (van der Leest et al. 2021). Han et al. obtained similar results from 33 patients with advanced NSCLC and detectable ctDNA levels; baseline ctDNA positivity had shorter OS and PFS (Han et al. 2022). Assaf et al. performed longitudinal ctDNA testing in the large randomized phase 3 IMpower150 study and found that higher OS and PFS were associated with low baseline ctDNA levels in a multivariate analysis (Assaf et al. 2023). Notably, Nabot et al. demonstrated that lung cancer patients could be effectively predicted to benefit from ICIs using a pretreatment composite model (DIREct-Pre), which includes tumor PD-L1 expression, pretreatment ctDNA, and circulating immune cell profiles (Nabot et al. 2020). In NSCLC, it is widely recognized that ctDNA is strongly associated with tumor load

(Abbosh et al. 2017; Chaudhuri et al. 2017; Avanzini et al. 2020). Zhang et al. comprehensively analyzed the ctDNA data from patients treated with durvalumab. The results showed that ctDNA was not only an alternative indicator of tumor burden, but that the correlation between ctDNA and OS remained after adjusting for smoking status, tumor burden, tumor PD-L1 score, baseline liver metastases, baseline lymph node metastases, and Eastern Cooperative Oncology Group (ECOG) performance status (Zhang et al. 2020). However, this study also found that ctDNA before treatment was not associated with the objective response rate (ORR) (Zhang et al. 2020).

In conclusion, pretreatment ctDNA level correlates with OS and other identified prognostic variables, but not with the ORR, and it may be a predictive biomarker in metastatic lung cancer receiving immunotherapy (Fig. 1).

ctDNA levels during treatment for predicting the response to ICIs in lung cancer

ctDNA kinetics measurement during early therapy

Tumors treated with ICIs has different response patterns than those treated with targeted therapy or chemotherapy. Some patients treated with ICIs experience transient enlargement due to immune cell infiltration before delayed tumor shrinkage, which is known as pseudoprogression (Chiou and Burotto 2015; Queirolo and Spagnolo 2017; Tazdait et al. 2018; Frelaut et al. 2020). Given this unique challenge, conventional radiologic response criteria do not identify all patients who benefit from ICIs (Eisenhauer et al. 2009; Ribas et al. 2009; Wolchok et al. 2009). Therefore, there is an

important need to determine the early response to ICIs using new biomarkers (Fig. 2). Some studies have demonstrated a relationship between ctDNA kinetics, tumor response, and outcomes in lung cancer patients receiving ICIs alone or combined with chemotherapy (Table 2). However, the time points and ctDNA change thresholds used in these studies to assess molecular responses varied. Therefore, determining the best cutoff threshold and time point for comparison with baseline ctDNA levels during ICI therapy is vital. In addition to driver mutations, ctDNA changes can also be measured by neoantigen-encoding mutations (Jia et al. 2020) and copy number aberrations (Weiss et al. 2017; Jensen et al. 2019).

In the early stage of ICI therapy, ctDNA changes correlate with disease progression as assessed by radiographic imaging, suggesting their ability to predict lung cancer response (Goldberg et al. 2018; Anagnostou et al. 2019; Guibert et al. 2019; Herbreteau et al. 2020). However, the

Table 1 Predictive value of pre-treatment ctDNA level in lung cancer immunotherapy

Cancer type	Sample size	Treatment	Technology	Group	Endpoint	Clinical/Trial NO.	References
Resectable stage IIIA NSCLC	n=43	Neoadjuvant chemotherapy+nivolumab	NGS	< 1% MAF vs. $\geq$ 1% MAF	PFS OS	NCT03081689	Provencio et al. (2022)
Unresectable stage IIB-IIIb NSCLC	n=218	Chemoradiation vs. chemoradiation+durvalumab/atezolizumab	CAPP-seq	Undetectable ctDNA vs. detectable ctDNA	FFP	NCT00349830 NCT02525757	Moding et al. (2020)
Metastatic NSCLC	n=15	Nivolumab Pembrolizumab	NGS, ddPCR, Bi-PAP	Undetectable ctDNA vs. detectable ctDNA	PFS OS	NCT02866149	Cabel et al. (2017)
Advanced lung adenocarcinoma	n=100	Nivolumab Pembrolizumab Atezolizumab Durvalumab	ddPCR	Low mutant copy levels vs. high mutant copy levels	PFS OS	N/A	Van der Leest et al. (2021)
Advanced NSCLC	n=33	Sintilimab+docetaxel	NGS	ctDNA positive ($\geq$ 2 mutations detectable) vs. ctDNA negative ($\leq$ 1 mutation detectable)	PFS OS	ChiCTR1900027634	Han et al. (2022)
Stage IV nonsquamous NSCLC	n=466	Atezolizumab+bevacizumab+carboplatin+paclitaxel Atezolizumab+carboplatin+paclitaxel Bevacizumab+carboplatin+paclitaxel	NGS	Below the median ctDNA level at baseline vs. above the median ctDNA level at baseline	PFS OS	NCT02366143	Assaf et al. (2023)
Stage IV NSCLC	n=99	Atezolizumab	CAPP-Seq	High DIREct-Pre score vs. low DIREct-Pre score	PFS	NCT01903993	Nabet et al. (2020)
Advanced NSCLC	n=978	Durvalumab Durvalumab+tremelimumab	NGS	Increased VAF vs. decreased VAF	OS ORR	NCT02008227 NCT01693562 NCT02087423 NCT02261220	Zhang et al. (2020)

NSCLC Non-small cell lung cancer; NGS next-generation sequencing; CAPP-seq cancer personalized profiling by deep sequencing; ddPCR droplet digital PCR; Bi-PAP bidirectional-phosphorolysis-activated polymerization PCR; MAF mutant allele fraction; DIREct-Pre Durable Immunotherapy Response Estimation by immune profiling and ctDNA- Pre-treatment; 1/AF variant allele frequency; FFP freedom from progression; PFS progression-free survival; OS overall survival; ORR objective response rate; N/A not available

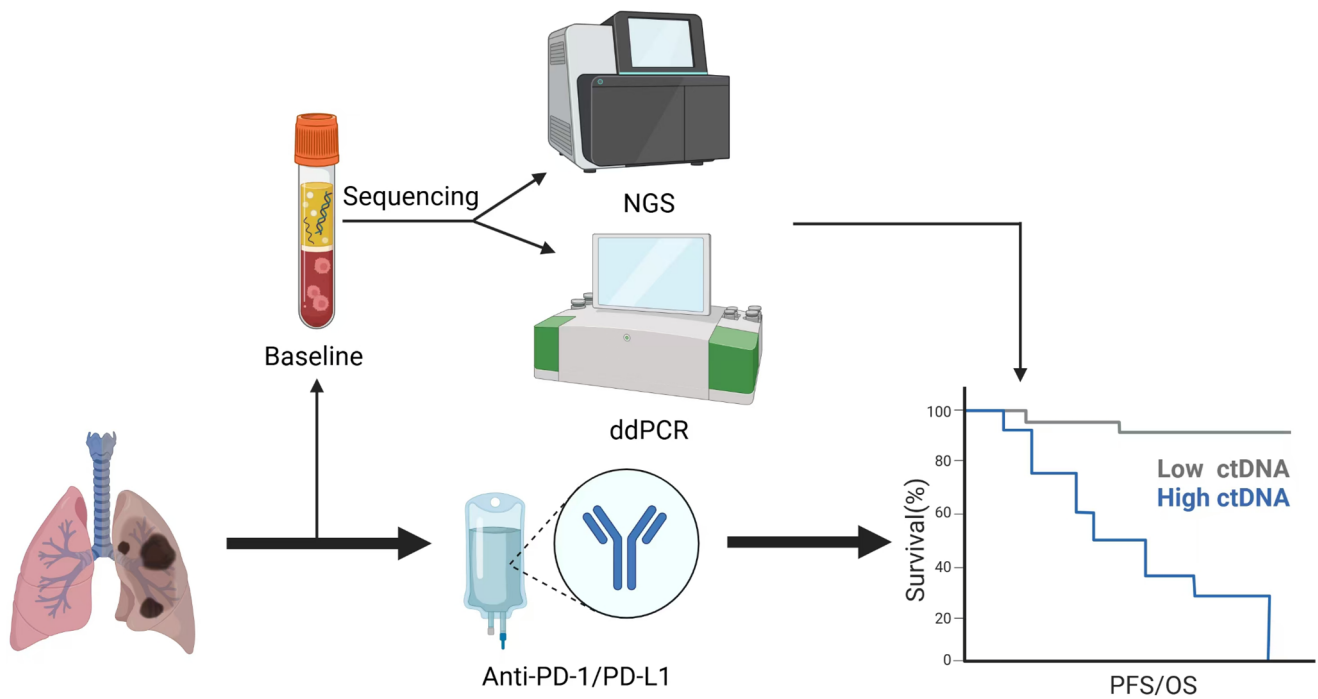


Fig. 1 Pretreatment levels of ctDNA for predicting the response to ICIs in lung cancer

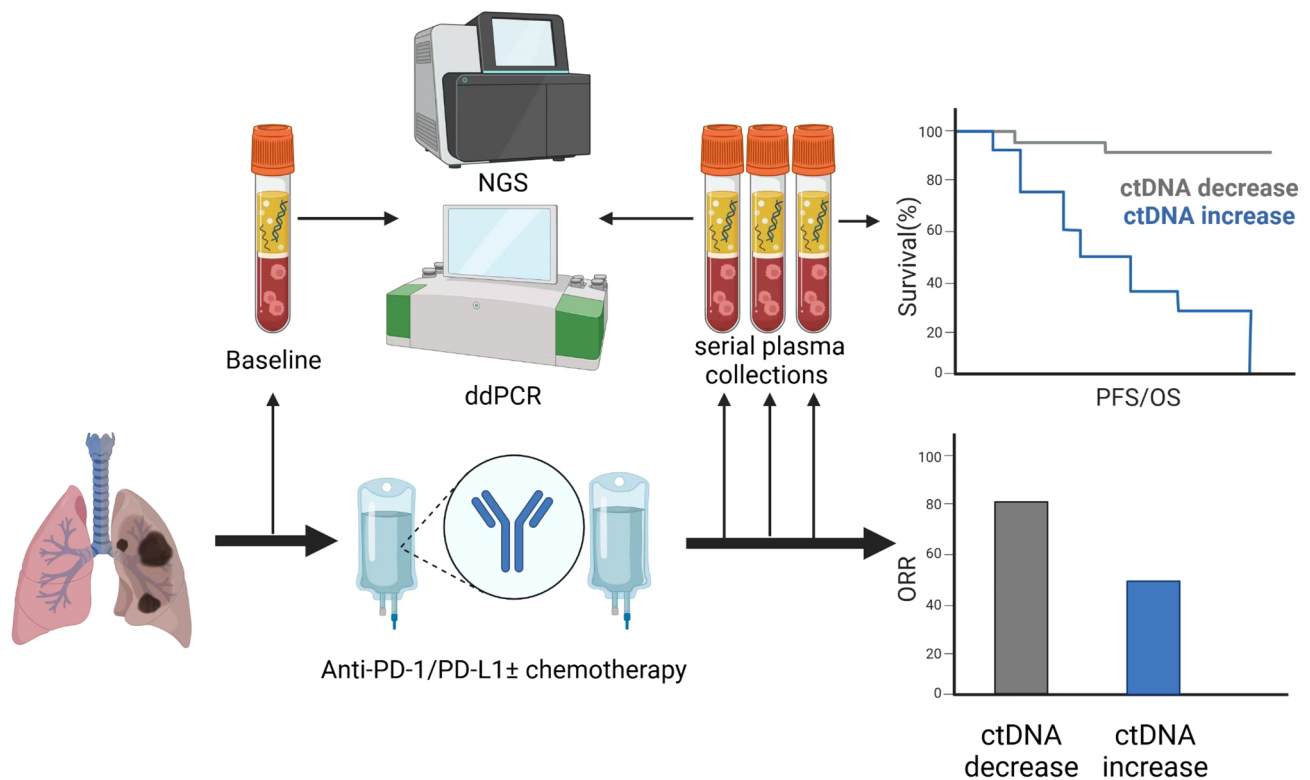


Fig. 2 ctDNA levels during treatment for predicting the response to ICIs in lung cancer

Table 2 Predictive value of on-treatment ctDNA level in lung cancer immunotherapy

Cancer type	Sample size	Treatment	Technology	Group	Endpoint	Clinical Trial NO.	References
Unresectable stage IIB-IIIB NSCLC	n=218	Chemoradiation vs. chemoradiation + durvalumab/atezolizumab	CAPP-seq	Increased ctDNA concentrations vs. Decreased ctDNA concentrations	FFP	NCT00349830 NCT02525757	Moding et al. (2020)
Metastatic NSCLC	n=15	Nivolumab Pembrolizumab	NGS, ddPCR, Bi-PAP	Undetectable ctDNA vs. detectable ctDNA	PFS OS	NCT02866149	Cabel et al. (2017)
Advanced lung adenocarcinoma	n=100	Nivolumab Pembrolizumab Atezolizumab Durvalumab	ddPCR	Increased ctDNA level vs. decreased ctDNA level	PFS OS	N/A	Van der Leest et al. (2021)
Stage IV nonsquamous NSCLC	n=466	Atezolizumab + bevacizumab + carboplatin + paclitaxel Atezolizumab + carboplatin + paclitaxel Bevacizumab + carboplatin + paclitaxel	NGS	ctDNA low risk (C3 ctDNA < 1 MTM) vs. ctDNA high risk (C3 ctDNA ≥ 1 MTM)	PFS OS	NCT02366143	Assaf et al. (2023)
Stage IV NSCLC	n=99	Atezolizumab	CAPP-Seq	> 50% decrease in ctDNA level vs. ≤ 50% decrease	PFS	NCT01903993 NCT02008227	Nabet et al. (2020)
Advanced NSCLC	n=171	Durvalumab Durvalumab + tremelimumab	NGS	Increased VAF vs. decreased VAF	PFS OS ORR	NCT01693562 NCT02087423 NCT02261220	Zhang et al. (2020)
Metastatic NSCLC	n=28	Nivolumab, pembrolizumab	NGS	≥ 50% decrease in ctDNA level vs. < 50% decrease	PFS OS	N/A	Goldberg et al. (2018)
Advanced NSCLC	n=20	Nivolumab	NGS	ctDNA <0.006 ng/μl vs. ctDNA >0.006 ng/μl	PFS OS	N/A	Giroux Leprieur et al. (2018)
Advanced NSCLC	n=28 (cohort 1) n=72 (cohort 2)	Durvalumab	NGS	Increased VAF vs. decreased VAF	PFS OS ORR	NCT01693562 NCT02087423	Raja et al. (2018)
Metastatic NSCLC	n=24	Nivolumab, pembrolizumab	NGS	Undetectable ctDNA vs. detectable ctDNA	PFS OS	N/A	Anagnostou et al. (2019)
stage IIB/IV NSCLC	n=97	Nivolumab	NGS	Increased AF vs. decreased AF	PFS	NCT02827344	Guibert et al. (2019)
SCLC	n=68	Atezolizumab vs. chemotherapy	NGS	below the median relative ctDNA abundance vs. above the median relative abundance	PFS OS	NCT03059667	Herbreteau et al. (2020)
Advanced NSCLC	n=62	Pembrolizumab ± chemotherapy	NGS	Decreased AF at first follow-up vs. increased AF at first follow-up	PFS OS ORR	N/A	Ricciuti et al. (2021)
Locally advanced or metastatic NSCLC	n=94	Atezolizumab vs. chemotherapy	NGS	High ctDNA above median MMPM vs. low ctDNA below median MMPM	OS	NCT02008227	Zou et al. (2021)
Stage IIB-IV squamous NSCLC	n=390	Camrelizumab + chemotherapy vs. placebo + chemotherapy	NGS	Uncleared ctDNA vs. cleared ctDNA	PFS OS	NCT03668496	Ren et al. (2022)

NSCLC Non-small cell lung cancer; SCLC Small cell lung cancer; CAPP-seq cancer personalized profiling by deep sequencing; NGS next-generation sequencing; ddPCR droplet digital PCR; Bi-PAP bidirectional-pyrophosphorolysis-activated polymerization PCR; MTM mean tumor molecules per ml plasma; VAF variant allele frequency; AF allele frequency; FFP freedom from progression; PFS progression-free survival; OS overall survival; ORR objective response rate; N/A not available

definition of ctDNA molecular response remains unclear. Any ctDNA reduction (Zhang et al. 2020), > 50% reduction in ctDNA levels (Goldberg et al. 2018; Moding et al. 2020; Nabet et al. 2020), and complete elimination (Anagnostou et al. 2019; Ren et al. 2022) have all been used to define ctDNA molecular responses and predict response to ICIs. It is worth mentioning that when ctDNA is eliminated, superior long-term clinical outcomes are observed in patients with advanced NSCLC, suggesting that it may also be a longitudinal predictor for accurately predicting durable clinical response (Anagnostou et al. 2019; Moding et al. 2020; Zhang et al. 2020).

Early assessment of ctDNA kinetics can distinguish lung cancer patients who eventually benefit from ICIs and patients who discontinue treatment because of serious side effects or those with stable disease at the first radiological evaluation (Anagnostou et al. 2019; Nabet et al. 2020; Zhang et al. 2020). However, the initial radiological evaluation of the tumor response may be inconsistent with the ctDNA molecular response (Ricciuti et al. 2021), possibly owing to pseudoprogression (Frelaut et al. 2020). Although a rare phenomenon (Ferrara et al. 2019), pseudoprogression remains a challenge in clinical practice. Monitoring ctDNA dynamics can help distinguish between pseudoprogression and true disease progression (Guibert et al. 2017; Ricciuti et al. 2021). In a case report describing three patients with lung adenocarcinoma during anti-PD-1 treatment, Guibert et al. observed that ctDNA levels showed a rapid and dramatic decrease in two patients with pseudoprogression, whereas they increased significantly in a patient with true progression (Guibert et al. 2017). In a retrospective analysis of 62 patients with advanced NSCLC, 17 had undetectable baseline ctDNA and only one had pseudo-progression, who received first-line pembrolizumab with or without platinum and pemetrexed (Ricciuti et al. 2021). However, these studies were conducted using limited or retrospective data, and their results should be further validated in larger cohorts.

ctDNA is another option for early endpoints in clinical decision-making. Recently, Assaf et al. divided patients with metastatic nonsquamous NSCLC receiving a combination of chemo-immunotherapy into training and test datasets in the phase 3 IMpower150 clinical trial. They then resampled the three treatment groups in the test dataset and, using various early endpoints assessed at week 6, evaluated the simulated.

proportion of treatment response rates that were higher in the active group than in the control group (Assaf et al. 2023). The results demonstrated that the true-go rate of ctDNA response was higher than that of RECIST or PFS at week 6, indicating that ctDNA response may be a more reliable indicator of drug efficacy than early radiologic assessment, suggesting that the ctDNA response could be more helpful

in predicting drug efficacy than early radiographic evaluations (Assaf et al. 2023). In addition, the true-go rate was enhanced by combining the ctDNA response metric with PFS in comparison to PFS alone, as well as by combining the ctDNA response metric with RECIST in comparison to RECIST alone (Assaf et al. 2023). Collectively, these findings suggest that ctDNA measurement during immunotherapy may be useful as an early endpoint to inform therapy decisions in clinical trials.

ctDNA dynamics

The change in ctDNA levels from pretreatment to during treatment may also directly predict the clinical outcome. Some studies have correlated clinical outcomes in NSCLC patients with ctDNA levels after the first cycle of ICI treatment, and the association was significantly superior to that of pretreatment ctDNA levels (Giroux Leprieur et al. 2018; Cabel et al. 2018; Zhang et al. 2020). However, whether ctDNA dynamics or levels during treatment are more accurate for predicting clinical efficacy remains unclear. Most ctDNA dynamics studies do not distinguish patients with low pretreatment ctDNA with a significant decrease during treatment from patients with a high pretreatment ctDNA and a smaller decrease. For this reason, Zhang et al. constructed a framework by combining variant allele frequency (VAF) data before and after treatment and found that the higher the ratio of treatment to pre-treatment, the more effective the immunotherapy (Zhang et al. 2020). Studies have demonstrated that patients with ctDNA clearance during treatment have significantly longer PFS and OS (Cabel et al. 2017; Anagnostou et al. 2019; Zhang et al. 2020). Therefore, using both pretreatment ctDNA levels and those during treatment to stratify patients and calculating ctDNA clearance can help determine which patients will benefit most from immunotherapy.

Although NSCLC patients treated with ICIs experience sustained responses, a large proportion of patients eventually progress, and early ctDNA dynamics are not the best indicator of long-term prognosis. By conducting an analysis of the ctDNA in the blood samples of 31 NSCLC patients collected at a median of 26.7 months after the initiation of therapy, Hellmann et al. discovered that at the surveillance timepoint, the four patients in whom ctDNA was detected eventually underwent disease progression, whereas 25 out of the 27 patients in whom ctDNA was not detected experienced progression-free (Hellmann et al. 2020). This finding confirmed that long-term survival correlates with ctDNA clearance at any stage of immunotherapy (Bratman et al. 2020). As the optimal duration of ICI treatment in responding patients remains uncertain, additional ctDNA evaluation in the late stages of treatment is helpful to better identify

patients who are likely to have long-term benefits. In this case, it is also important to improve the sensitivity of ctDNA detection methods to reduce false-negative results.

Using ctDNA to assess genetic predictors of ICI therapy response

Blood-derived tumor mutation burden

TMB is the number of somatic nonsynonymous mutations or all mutations per megabase in the gene region detected in a tumor sample (Sha et al. 2020). These somatic mutations can lead to the formation of neoantigens on the surface of tumor cells, thereby triggering anti-tumor immune responses (McGranahan et al. 2016). According to some research, patients with high tissue-derived TMB (tTMB) and advanced NSCLC obtain greater benefits from ICI treatment (Rizvi et al. 2018; Ready et al. 2019). However, tTMB measurement may be limited by high clonal complexity and cannot be fully captured by a single biopsy; low tumor purity also affects its accuracy (Anagnostou et al. 2020). Blood-derived TMB (bTMB) determination has been evaluated as a surrogate for tissue-derived TMB. Several studies have shown a concordance between bTMB and tTMB (Koeppel et al. 2017; Chaudhuri et al. 2017; Yang et al. 2018; Gandara et al. 2018), and bTMB has been proposed as an emerging biomarker for predicting ICI treatment response (Rizvi et al. 2020; Kim et al. 2022).

Several clinical trials have evaluated the predictive role of bTMB on ICI response (Fig. 3). In a retrospective analysis of the POPLAR and OAK trials, Gandara et al. found that patients with a high bTMB had higher PFS and OS than those patients with a low bTMB (Gandara et al. 2018). A bTMB ≥ 16 mutations/Mb has been proven to be a technically reliable and clinically significant cutoff in NSCLC (Gandara et al. 2018). Similarly, in the MYSTIC randomized clinical trial, a bTMB ≥ 20 mutation/Mb was associated with better OS, PFS, and ORR compared to chemotherapy in patients with advanced NSCLC treated with durvalumab and tremelimumab (Rizvi et al. 2020).

Although several studies have highlighted the potential of bTMB to predict ICI response, this was not confirmed in a prospective analysis using pre-specified bTMB thresholds. A bTMB ≥ 16 mutations/Mb was associated with a better ORR in the phase 2 B-FIRST trial, which enrolled patients with locally advanced or metastatic NSCLC treated with atezolizumab, and bTMB thresholds were associated with higher ORR (Kim et al. 2022). However, there was no significant difference in PFS between the high and low bTMB subgroups when the predetermined cutoff of bTMB ≥ 16 mutations/Mb was used (Kim et al. 2022). B-FAST was

the first prospective phase 3 randomized trial to evaluate bTMB as a biomarker for predicting ICI response (Peters et al. 2022). The results demonstrated that the difference in PFS between the atezolizumab arm and the chemotherapy arm was not statistically significant in NSCLC patients with a bTMB ≥ 16 mutations/Mb (Peters et al. 2022). The exploratory analysis of this experiment also showed that the predictive ability of the bTMB increases with an increase in the threshold, which is consistent with the results from the B-FAST trial (Peters et al. 2022). Similarly, the primary endpoint of improving OS in patients with metastatic NSCLC and a bTMB ≥ 20 mutations/Mb did not reach in the phase 3 NEPTUNE trial comparing durvalumab plus tremelimumab with chemotherapy (AstraZeneca 2019; De Castro et al. 2023).

It is evident from these trials that the clinical utility of bTMB as a biomarker of response to ICIs in NSCLC patients is unclear. In addition to the lack of standardized detection techniques, these differences among the study results may be caused by the genes sequenced, the mutations analyzed, the allele frequency cutoff used for selection, and the threshold used to define a high bTMB (Sinoquet et al. 2023). To this end, using allele frequency-adjusted bTMB, Wang et al. demonstrated that only a high bTMB was associated with improved OS, PFS, and ORR in NSCLC patients who received ICI treatment (Wang et al. 2020). Similarly, Nabet et al. defined normalized bTMB as the ratio of bTMB to ctDNA levels. In their study, normalized bTMB predicted a durable clinical benefit in ICI-treated NSCLC patients and was better than the two metrics alone (Nabet et al. 2020).

Specific cancer mutations in ctDNA

Specific mutations in ctDNA can also be used to predict ICI resistance in NSCLC (Rizvi et al. 2018; Marinelli 2020). Guibert et al. found that transversion mutations, the substitution of a purine for a pyrimidine (Ruppert et al. 2020), in *TP53* and *KRAS* had a better prognosis, whereas the existence of *PTEN* or *STK11* mutations in ctDNA could cause early progression (Guibert et al. 2019). This may be due to *STK11* mutations were associated with a non-inflamed tumor microenvironment and reduced PD-L1 expression (Koyama et al. 2016), and loss of *PTEN* resulted in decreased T-cell infiltration and increased immunosuppressive cytokines expression in tumor microenvironment, as well as poor outcomes under ICI therapies (Peng et al. 2016). While *KRAS* and *TP53* mutated tumors showed prominently increased PD-L1 expression in tumor microenvironment and significantly increased mutation burden, which enriched in a transversion-high cohort (Dong et al. 2017). Therefore, combined with the association between known gene mutations and good or bad outcomes, they built an algorithm to

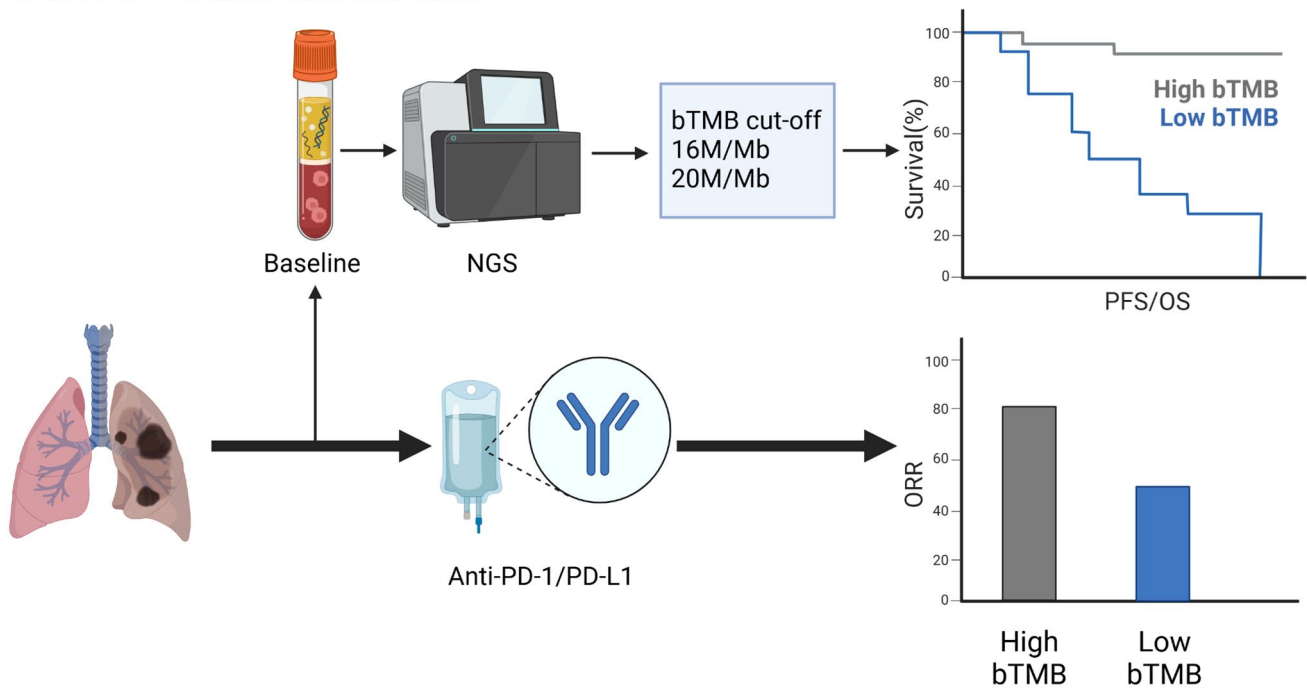
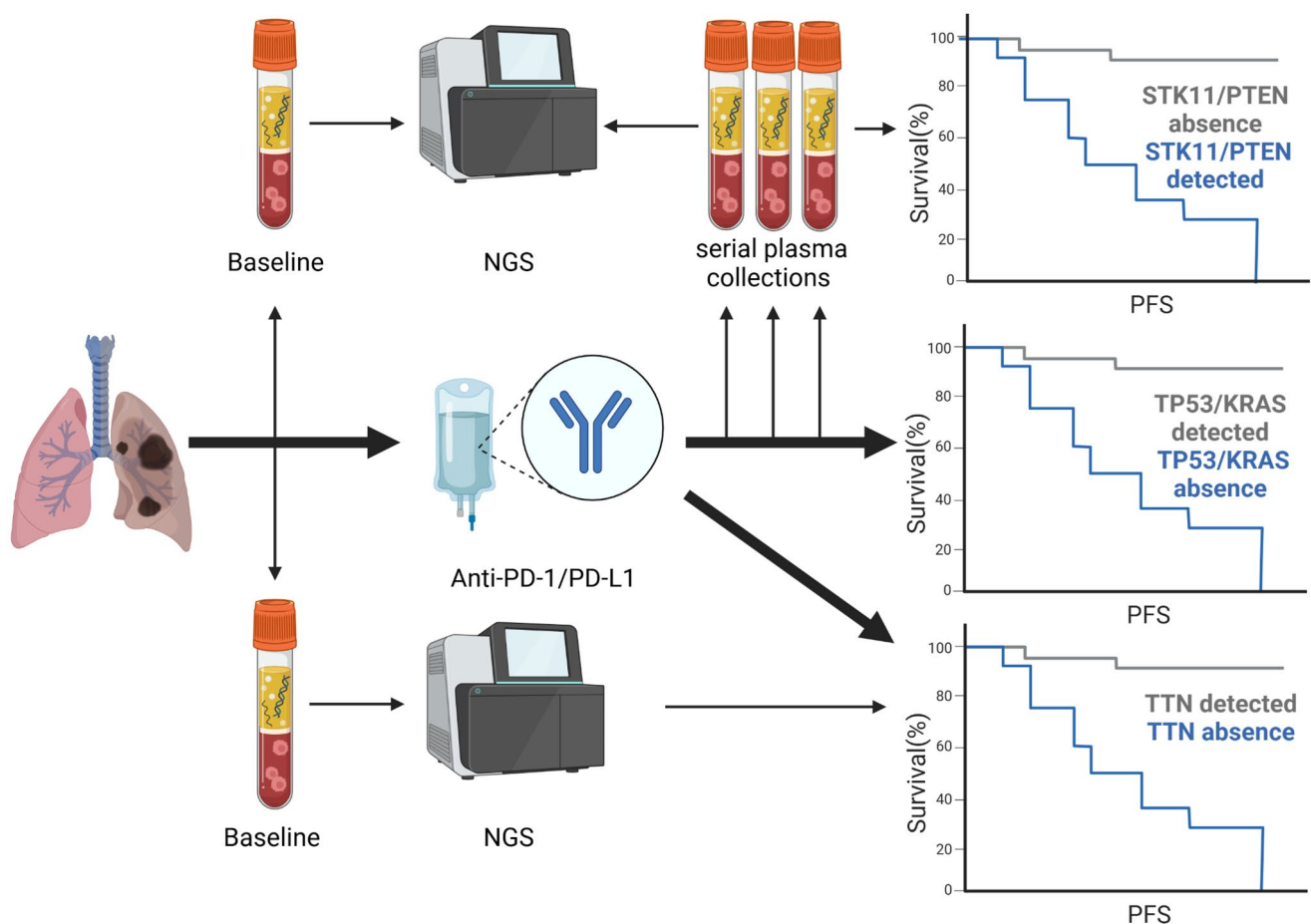
a. Blood-derived tumor mutation burden**b. Specific cancer mutations**

Fig. 3 Using ctDNA to assess genetic predictors of ICI therapy response. **a.** blood-derived tumor mutation burden. **b.** specific cancer mutations in ctDNA

divide patients into high or low immune score groups and determined who would most benefit from ICIs (Guibert et al. 2019). Similarly, in 66 patients with metastatic NSCLC receiving pembrolizumab, a bTMB ≥ 16 mutations/Mb and lack of *STK11*, *Keap1*, *PTEN*, or *ERBB2* exon 20 insertion mutations had better PFS and OS (Aggarwal et al. 2020). Titin (*TTN*) mutations were associated with a high TMB, high ORR, and durable survival in seven clinical cohorts, including patients with NSCLC treated with ICIs (Jia et al. 2019). Further research on ctDNA sequencing of *TTN* mutations in the peripheral blood showed that the clinical efficacy of ICI therapy in patients with NSCLC could be predicted by ctDNA *TTN* mutations (Su et al. 2021). *TTN* emerged as the gene possessing the longest coding sequence within the human genome. The highest quantity of mutations was identified in its coding regions, thereby suggesting a significant probability of accumulating somatic mutations in this gene (Su et al. 2021). In addition, deleterious mutations in ctDNA PTPRD could be used as prognostic biomarkers to predict longer PFS and greater ORR in nonsquamous NSCLC patients receiving ICI treatment (Sun et al. 2021).

Specific mutations associated with ICI resistance can be identified using large gene panels or WES. After sequential sequencing of ctDNA from 12 NSCLC patients, mutations in *PITCH1* and *B2M* were found in two of four patients with disease progression (Li et al. 2019). WES of ctDNA from eight NSCLC patients revealed the molecular mechanisms, including loss of *PTEN* or *B2M*, which played a role by inducing the production of immunosuppressive cytokines or impairing the processing and presentation of HLA class I antigen, correlated with late progression after ICI therapy (Giroux Leprieur et al. 2020). Considering the impact of sample size on the results, we should validate in a larger cohort of patients with lung cancer and further explore the molecular status of the tumor by ctDNA analysis at the onset of ICI treatment resistance (Fig. 3).

Limitations and future directions

Numerous studies have proven that ctDNA detection and kinetic evaluation can evaluate the response to ICIs at different periods of lung cancer treatment; therefore, they have gradually become promising predictive and prognostic biomarkers. However, there are many hurdles to overcome before these methods can be applied in clinical cancer care.

Currently, the detection of ctDNA is a crucial factor but limits its widespread application in the prediction of immunotherapy for lung cancer. There is no standardized detection method for ctDNA, and different detection methods have different fragment sizes, detection limits, and coverage depths (Cristiano et al. 2019). The sensitivity of ctDNA

analysis needs to improve before it can be fully incorporated into clinical practice. Additionally, the sample sizes in the majority of studies are relatively small, and the efficacy and universality of the ctDNA detection techniques are restricted.

Despite these challenges, ctDNA can also be used as a candidate to evaluate genetic predictors of ICIs responses, such as TMB. However, the clinical response to ICIs depends on the intricate interactions of the cancer-immune system and is driven by tumor-intrinsic and extrinsic features that mediate immunoediting mechanisms (Anagnostou et al. 2022). Therefore, multicomposite biomarkers integrating multiple indicators may most accurately predict the tumor response to ICIs. Blood-based circulating tumor cells, circulating cytokines, and circulating T cell subsets can provide information about multiple parameters of the tumor immunotherapy response and can be used as separate biomarkers of clinical efficacy (Guibert et al. 2018; Janning et al. 2019; Yamauchi et al. 2021). Recently, Nabet et al. integrated pretreatment changes in tumor PD-L1 expression, normalized bTMB, percentage of circulating CD8 T-cells, and ctDNA dynamics early in treatment and developed the DIREct-On score to predict the clinical outcome of NSCLC patients treated with ICIs. In a multivariate Cox proportional hazards model that included a line of therapy, ECOG performance status, and age, this score was the only variable independently associated with PFS. Its clinical classification accuracy and prognostic value were also much higher than those of each indicator alone (Nabet et al. 2020). ctDNA can also be used to analyze methylation patterns (Keller et al. 2021). An epigenomic profile based on a microarray DNA methylation signature (EPIMMUNE) was created in a sample of 34 NSCLC patients receiving pembrolizumab or nivolumab. This profile was then validated in another set of patients. The findings demonstrated a positive relationship between PFS/OS and EPIMUNE signature (Duruiseaux et al. 2018). Therefore, ctDNA analysis of multiple tumor characteristics can help develop biomarkers to predict response to ICIs and can be performed in a minimally invasive manner.

In the future, guidelines and best practices for ctDNA evaluation must be developed, which will help improve comparability between studies. Moreover, interventional clinical trials based on ctDNA are needed to prove its clinical utility as a biomarker. Further, the ctDNA detection threshold may vary depending on the time point and the detection method used. Thus, it may be more accurate to determine ctDNA changes or clearance at more consecutive time points rather than a single time point. Although most studies detect ctDNA at only one specific time point, more complex models can stratify patients by evaluating ctDNA from multiple samples collected at any one point in time in

the future. Additionally, regarding ICI therapies combined with other treatments, whether ctDNA still possesses predictive efficacy is also an aspect that requires attention in the future.

Many clinical studies have confirmed that ctDNA detection and its kinetic evaluation can evaluate ICI treatment response at different stages of lung cancer and have predictive and prognostic value. In the future, we need to explore whether ctDNA analysis can bring more clinical benefits to patients with lung cancer in larger studies, and define the ctDNA cutoffs in different clinical settings, such as pretreatment, early treatment, and disease progression. Moreover, the continuous development of ctDNA detection technology should focus on improving efficiency, accuracy, and cost.

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Data availability No datasets were generated or analysed during the current study.

Declarations

Ethics approval and consent to participate Not applicable.

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